

Profile Summary

- Data Engineer with **4.5+ years** of experience designing and optimizing **ETL/ELT pipelines**, **cloud-ready data architectures**, and **distributed processing solutions** supporting large-scale analytics across **finance**, **automotive**, and enterprise domains.
- Skilled in **Python, SQL, Spark, Airflow, Snowflake**, and **cloud platforms** (AWS, Azure, GCP), delivering reliable, high-performance **data ingestion workflows** that reduce latency and improve reporting efficiency.
- Experienced in **data modeling**, **data warehousing**, **orchestration**, and **real-time processing**, building automated, **fault-tolerant pipelines** that power **BI dashboards**, **advanced analytics**, and operational insights.
- Strong collaborator with cross-functional teams, translating business needs into **scalable data solutions** while ensuring **data quality**, **governance**, **validation**, **monitoring**, and **CI/CD integration**.

Core Technical Skills

Programming & Scripting: Python (Pandas, PySpark), SQL, Scala, Java, Shell, Go, Node.js, PL/SQL
Big Data & Streaming: Apache Spark, PySpark, Spark Streaming, Hadoop, Kafka, Kinesis, Flink, Hive, NiFi
Databases: PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, DynamoDB
Cloud Platforms:

- AWS:** S3, Glue, EMR, Redshift, Lambda, Kinesis, Athena, CloudWatch
- Azure:** Data Factory, Data Lake, Synapse, Databricks

Data Warehousing: Snowflake, Redshift, Teradata, Synapse
Orchestration & DevOps: Airflow, dbt, Jenkins, Git, Docker, Kubernetes, Terraform, CI/CD
Visualization: Tableau, Power BI, Looker
Other: Databricks (MLflow), REST/GraphQL APIs, Data Governance (GDPR, HIPAA)

Professional Experience:

Software Engineer | General Motors, Michigan, USA | Jan 2024 – Present

- Improve **diagnostic data stability** and workflow efficiency by enhancing **automation frameworks** across large-scale vehicle testing processes.
- Built **Python-based validation tools** to improve **data accuracy** and reduce rework, increasing reliability across multiple vehicle variants.
- Integrated automated checks into **Jenkins CI/CD pipelines**, reducing manual verification efforts by **35%** and boosting delivery efficiency.
- Developed **interactive dashboards** using **Python** and **Power BI**, enabling real-time tracking of key diagnostic performance metrics.

Data Engineer | Emonics LLC, New Jersey, USA | Aug 2023 – Dec 2023

- Built **scalable ETL pipelines** using **Spark** and **Airflow**, **processing 500GB+** daily data for enterprise analytics.
- Designed AWS workflows** using S3, Glue, Redshift, cutting data latency **30%** and improving reporting availability.
- Built scalable data models** and validation logic, **improving BI accuracy**, audit readiness, and consistent production data quality.

Data Engineer Intern | IMPEL, New York, USA | Jan 2023 – May 2023

- Developed ingestion and transformation pipelines** improving data consistency, reducing work, accelerating analytics readiness.
- Optimized Spark jobs**, achieving **40%** faster runtime performance across distributed clusters handling large-scale datasets.
- Built Redshift data marts centralizing analytics, enabling faster dashboards, improved reporting, and data reusability.

Data Engineer | American Express, Bengaluru, India | Jun 2019 – Dec 2021

- Developed reliable ETL pipelines with Python, SQL, and AWS Glue**, processing high-volume financial data accurately.
- Improved **Redshift query performance 25%** via tuning, partitioning, and data model enhancements, boosting analytics speed.
- Built **secure, traceable data pipelines supporting fraud detection**, compliance reporting, and enterprise risk management initiatives.

Education:

Masters in Business Analytics – Trine University, Michigan | USA (2026)
Masters in Data Science – University at Buffalo, New York (2023)
Bachelors in Computer Science – Gujarat Technological University, India (2020)

Academic Projects:

- Reducing Commercial Aviation Fatalities |** Python-Pandas, NumPy, matplotlib, Scikit-learn, SQL
- Invented machine learning models to estimate real-time state of mind of pilots based on physiological data and alerting pilots when enter dangerous cognitive states and preventing potential accidents by more than 80%.
 - Executed Exploratory Data Analysis (EDA) to clean and analyse data, and applied 4+ models, including Logistic Regression, Decision Tree, Random Forest, and Ad boost Classifier, to achieve high scores.
- Face Recognition System for Boarding control at airport |** PIL, OpenCV, PyCharm, Pillow, MySQL, XAMPP Server
- Engineered a biometric information processing system that combines Face Detection and Face Recognition techniques in 2 stages, with the goal of accurately identifying passengers boarding flights.
 - Operationalized logic to apply Face Detection to live acquired images and Face Recognition algorithms for image processing. The system then checks passenger’s eligibility to board based on the results.